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FIRST SEMESTER B.C.A. (REVISED NEP) DEGREE
EXAMINATION, JANUARY/FEBRUARY 2025
INTRODUCTION TO ALGORITHMS
Theory

Time : 3 Hours] [Max. Marks : 80

Instructions : *1) Part – **A** : Answer **all ten** questions, **each** carries **two** marks.*

*2) Part – **B** : Answer **any six** questions, **each** carries **five** marks.*

*3) Part – **C** : Answer **any three** questions, **each** carries **ten** marks.*

PART – A (10×2=20)

1. Define algorithm.
2. What is program ?
3. What is sorting ? Give an example for sorting technique.
4. What is worst case behaviour of algorithm ?
5. What is fibonacci number ?
6. Write an abbreviation of ASCII.
7. Give two examples for divide and conquer technique.
8. Write a pseudo code to swap two numbers.
9. Write any two applications of algorithm.
10. What is binary search ?

PART – B (6×5=30)

11. Explain the characteristics of algorithm.
12. What is top down design ? Write the advantage.
13. Write an algorithm for factorial computation.

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14. Explain sorting by selection.

15. Write a C program to find GCD of two numbers.

16. Write an algorithm to find smallest divisor of integer.

17. Explain the linear search technique with example.

18. Perform base conversion for the following :

    a) $(1101)_2$ convert to decimal number.

    b) $(973)_{10}$ convert to octal number.

(3×10=30)

PART – C

19. Explain implementation of algorithm.

20. Explain mergesort technique with example.

21. Write an algorithm for removal of duplicate elements from an array and explain with example.

22. a) Write a C program to find a square of a number without using build in function.

    b) Write an algorithm to generate prime numbers.
